

SOLARKITS ERP

Enterprise Client Documentation & Master Workflow

Project Name: SOLARKITS ERP System (2026 Edition)

Target Audience: Client Executive Leadership, System Administrators, & Engineering Audit Teams

Classification: Confidential Master Technical & Functional Specification

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1. Executive Summary & System Vision

SOLARKITS ERP is a state-of-the-art enterprise resource planning platform engineered specifically for the solar energy sector. It unifies complex operational workflows—spanning multi-level geographical territory management, tier-1 component supply chains, milestone-based customer financing, and strict multi-stage solar project construction—into a single, high-fidelity web application.

SOLARKITS Enterprise Ecosystem Overview			
Geography & Foundation	HR	Core Project Mgt (8 Stages)	Supply Chain & Finance
6-Level Regions		Lead & Survey	Warehouse & BOM
Dept Org Chart		Documentation	Tier-1 Brands
Roles & Access		Dispatch / Install	Custom ComboKit
Approvals / SLA		Net Metering	Loan Milestones

2. Comprehensive Technology Stack & Architecture

The platform is engineered on an optimized MERN (MongoDB, Express.js, React, Node.js) stack designed for sub-second UI responsiveness, secure role-based access control, and highly scalable data relationships.

2.1 Frontend Architecture (solar-client)

- **Core Framework:** React 18 with Vite build engine for near-instant hot module replacement and highly optimised production bundling.
- **Styling & Design System:** Vanilla CSS paired with Tailwind CSS utility classes, utilising modern glassmorphism, responsive CSS flex/grid layouts, and curated colour palettes.
- **Iconography:** lucide-react for clean, professional, and consistent vector SVG icons across all sidebar menus and KPI cards.
- **State Management:** Custom React hooks (useLocations, useAuth) combined with Zustand/Context for global session and permission persistence.
- **Spreadsheet Processing:** xlsx library for client-side Excel parsing, enabling high-speed bulk data uploads with instant client-side validation previews before server submission.

2.2 Backend Server Architecture (solar-server)

- **Server Runtime:** Node.js v20+ executing asynchronous Express.js REST API handlers.
- **Database & ORM:** MongoDB Enterprise Cloud managed via Mongoose ODM schemas, featuring strict field validation, compound geographical indexing, and cascading virtual joins.
- **Authentication & Security:** JSON Web Tokens (JWT) stored securely with bcrypt password hashing. Middleware enforces strict role-based access control (authorize('admin', 'dealer')) and department-level module delegation.
- **Data Aggregation:** Advanced Mongoose \$facet, \$lookup, and \$group aggregation pipelines to generate high-performance financial P&L, inventory valuations, and regional counts.

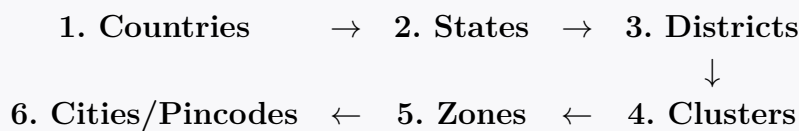
3. Exhaustive Breakdown of All 18 Core Modules

Master Navigation Tree

1. Dashboard (Orders, Loans, Installers, Delivery, Vendors, Inventory)
2. Departments (Organization Chart)
3. Approvals (Approval Summary & Requests)
4. Project Management (Company, Partners, Installer Agency)
5. Operations (Warehouse, Add Stock, Inventory Mgt, Brand Supplier)
6. Settings (Locations, HR, Vendor, Sales, Marketing, Delivery, etc.)
7. Reports (Financial P&L, Cashflow, Inventory, Loans, Captable, etc.)

3.1 Module 1: Location Settings & Hierarchy Management

6-Level Location Hierarchy



3.1.1 A. Business Purpose

The geographical foundation of the ERP. Establishes a 6-tier cascading territory tree used to filter projects, assign regional dealer leads, apply state-specific pricing markups, and calculate logistics transport freight.

3.1.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Location Settings → Setup Locations

URL: /admin/settings/location/setup-locations

1. **Countries Tab:** Dropdown of master global countries. Button: *Activate Country*.
2. **States Tab:** Table of active states. Form: State Name, Code, and Description.
3. **Districts Tab:** Table linking Districts to parent States. Form: District Name.
4. **Clusters Tab:** Multi-select grouping of several districts into a commercial cluster.
5. **Zones Tab:** Multi-cluster zones for senior territory managers.
6. **Cities Tab:** City management classified by Area Type (*Urban* vs *Rural*). Includes manual input and **Bulk Excel Upload** with instant table preview and internal duplicate filtering.

3.1.3 C. Backend API Endpoints & Mongoose Models

Models: Country.js, State.js, District.js, Cluster.js, Zone.js, City.js
(solar-server/models/core/)

- **GET** /api/locations/hierarchy/states?countryId=... — Returns cascading state tree.
- **POST** /api/locations/cities/bulk-create — Parses Excel JSON stream, validates unique city names and pincodes, executes MongoDB batch insert.

3.2 Module 2: Departments Module

3.2.1 A. Business Purpose

Defines the internal organisational chart of the solar company (e.g., Solar Engineering, Field Operations, Accounting, Sales). Controls which system modules each department can access.

3.2.2 B. UI Navigation & Sub-sections

Path: Sidebar → Departments → Organization Chart

URL: /admin/departments/organization-chart

1. **Department Master List:** Table of active departments showing assigned module badges. Button: *+ Add New Department*.
2. **Organization Chart View:** Visual reporting tree illustrating parent-child department relationships.

3.2.3 C. Backend API Endpoints & Mongoose Models

Model: Department.js (solar-server/models/hr/Department.js)

- **POST** /api/departments — Validates unique department code, creates Mongoose document.

- **GET** /api/departments/hierarchy — Returns nested JSON tree for React D3/Canvas organisation chart rendering.

3.3 Module 3: HRMS & Role Management Module

3.3.1 A. Business Purpose

Manages employee onboarding, candidate evaluation testing, job vacancy postings, leave/resignation requests, and granular role-based permissions across the company.

3.3.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → HR Settings (Expandable menu)

1. **Role Settings** (/admin/settings/hr/role-settings): Create designations (e.g., “Senior Field Engineer”, “Franchisee Manager”) and toggle module checkmarks.
2. **Create Department** (/admin/settings/hr/create-department): Direct department setup modal.
3. **Manage Employees** (/admin/settings/hr/manage-employees): Table of staff. Form: Name, Email, Password, Department, Role, and Reporting Manager.
4. **Department-wise Modules** (/admin/settings/hr/department-wise-modules): View and delegate module access across departments.
5. **Temporary Incharge Setting** (/admin/settings/hr/temporary-incharge-setting): Assign temporary managerial roles when department heads are on leave.
6. **Leave Approvals** (/admin/settings/hr/leave-approvals): Review employee leave applications. Buttons: *Approve* / *Reject*.
7. **Resign Approvals** (/admin/settings/hr/resign-approvals): Review employee resignation notices and exit interview logs.

3.3.3 C. Backend API Endpoints & Mongoose Models

Models: User.js, Role.js, Permission.js, LeaveRequest.js (solar-server/models/users/ & hr/)

- **POST** /api/users/register — Hashes password with bcrypt, commits employee record.
- **PUT** /api/hr/leaves/:id/status — Updates leave application state.

3.4 Module 4: Approvals & Overdue Settings Module

3.4.1 A. Business Purpose

Defines automated escalation thresholds and multi-level approval chains for high-value financial transactions, timeline extensions, and sensitive project changes.

3.4.2 B. UI Navigation & Sub-sections

Path: Sidebar → Approvals (/admin/approvals) or Sidebar → Settings → Overdue Setting

1. **Approval Dashboard** (/admin/approvals): KPI summary cards (Total Pending, Onboarding Approved, User Approved). Tabbed table of incoming approval tickets. Buttons: *Approve* / *Reject* with mandatory notes modal.
2. **Approval Overdue Setting** (/admin/settings/approval-overdue): Configure escalation timers for unaddressed approval tickets.
3. **Overdue Task Setting** (/admin/settings/overdue-task): Define **Pending Min/Max Days** (e.g., 1 to 7 days) and multi-level **Escalation Levels** (e.g., Day 3 overdue escalates to District Manager with a 2% penalty).
4. **Overdue Status Setting** (/admin/settings/overdue-status): Map specific overdue rules to geographical regions and employee roles.

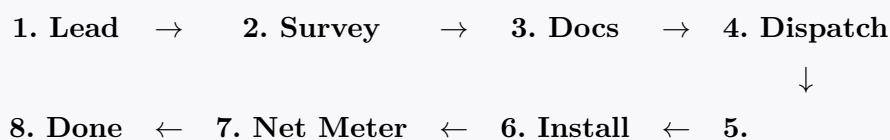
3.4.3 C. Backend API Endpoints & Mongoose Models

Models: Approval.js, OverdueTaskSetting.js, OverdueStatusSetting.js (solar-server/models/approvals/)

- **POST** /api/approvals/request — Creates an approval ticket, flags target entity status as *Under Approval*.
- **GET** /api/approvals/pending — Retrieves active tickets filtered by the logged-in administrator's permission level.

3.5 Module 5: Project Management Module (Core)

8 Strict Project Stages Flow



3.5.1 A. Business Purpose

The operational heart of SOLARKITS. Enforces a strict, unskippable 8-stage construction lifecycle for Residential, Commercial, and Agricultural solar installations.

3.5.2 B. UI Navigation & Sub-sections

Path: Sidebar → Project Management → Company / Partners / Installer Agency

1. **Management Projects** (.../management): Kanban and grid views of active projects. Clicking a project card opens the stage inspection modal. Form: Document uploads (Electricity bill, roof photos), customer sign-offs, and *Move to Next Stage* button.

2. **Install Projects** (.../install): Dedicated view for field engineers showing Stage 6 (Installation) active projects with mounting structure and wiring schematics.
3. **Service Projects** (.../service): Warranty and maintenance project tracking.
4. **Track Service Projects** (.../track-service): Live SLA countdown timers for urgent customer maintenance requests.

3.5.3 C. Backend API Endpoints & Mongoose Models

Models: `Project.js`, `StageHistory.js`, `ProjectChecklist.js` (solar-server/models/projects/)

- **GET** /api/projects — Fetches active projects. Includes query parameters for stage, location, and overdue status.
- **PUT** /api/projects/:id/stage — Verifies mandatory stage completion checklist, updates `project.currentStage`, records timestamp in `StageHistory`, recalculates overdue SLA timers.

3.6 Module 6: Operations & Warehouse Module

3.6.1 A. Business Purpose

Manages physical warehouse inventory, tracks real-time stock levels of solar kits, panels, inverters, and mounting structures, and processes material dispatch requests.

3.6.2 B. UI Navigation & Sub-sections

Path: Sidebar → Operations → Our Warehouse / Add Inventory Request / Inventory Management

1. **Our Warehouse** (/admin/operations/warehouse): Live inventory stock table. SKUs falling below minimum thresholds are flagged with a red “**Low Stock Alert**”.
2. **Add Inventory Request** (/admin/operations/add-inventory): Stock intake form. Input: SKU, batch number, supplier vendor, unit price, and received quantity.
3. **Inventory Management** (/admin/operations/inventory-management): Stock reconciliation, adjustments, and inter-warehouse transfer logs.

3.6.3 C. Backend API Endpoints & Mongoose Models

Models: `Warehouse.js`, `InventoryItem.js`, `StockTransaction.js` (solar-server/models/inventory/)

- **POST** /api/inventory/stock-in — Adds received quantity to `availableStock`, records audit log.
- **POST** /api/inventory/dispatch — Deducts stock upon Project Stage 4 approval, links dispatch invoice to Project ID.

3.7 Module 7: Brand Manufacturer & Supplier Module

3.7.1 A. Business Purpose

Maintains an exhaustive catalogue of approved tier-1 solar component manufacturers (e.g., Waaree, Vikram Solar, Sungrow, Growatt) and their wholesale supplier networks.

3.7.2 B. UI Navigation & Sub-sections

Path: Sidebar → Operations → Brand Manufacturer (Expandable menu)

1. **Add Brand Manufacturer** (/admin/operations/brand/add-brand-manufacturer): Form to onboard brand name, corporate logo, country of origin, warranty terms, and component category (Panels, Inverters, Batteries).
2. **Brand Supplier Overview** (/admin/operations/brand/supplier-overview): Analytical table comparing supplier delivery times, component defect rates, and credit terms.

3.7.3 C. Backend API Endpoints & Mongoose Models

Models: Brand.js, SupplierOverview.js (solar-server/models/inventory/& vendors/)

- **GET** /api/brands/active — Fetches active brand list to populate ComboKit customisation dropdowns.

3.8 Module 8: Vendor Management Module

3.8.1 A. Business Purpose

Manages third-party vendors including external installation contractors, logistics transport providers, and component distributors.

3.8.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Vendor Settings (Expandable menu)

1. **Installer Vendors** (/admin/settings/vendor/installer-vendors): Onboard third-party field contractors. Form: PAN, GSTIN, bank details, and service territory allocation (States/Districts).
2. **Supplier Type** (/admin/settings/vendor/supplier-type): Define vendor classifications (Hardware Distributor, EPC Contractor, Logistics Carrier).
3. **Supplier Vendors** (/admin/settings/vendor/supplier-vendors): Master table of all active wholesale component suppliers.

3.8.3 C. Backend API Endpoints & Mongoose Models

Model: Vendor.js (solar-server/models/vendors/Vendor.js)

- **POST** `/api/vendors` — Validates GST format, creates vendor document, indexes geographical coordinates.

3.9 Module 9: Installer Agency & Settings Module

3.9.1 A. Business Purpose

Manages professional solar installation teams and agencies. Tracks their certification ratings, physical tool requirements (e.g., crimping tools, safety harnesses), and project allocation limits.

3.9.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Installer Settings (Expandable menu)

1. **Solar Installer** (`/admin/settings/installer/solar-installer`): Profiles of field technicians. Displays completed project counts and average customer star ratings.
2. **Installer Tool Requirements** (`/admin/settings/installer/tool-requirements`): Mandatory safety and electrical tool checklists. Field technicians must upload photos during annual audits.
3. **Rating Setting** (`/admin/settings/installer/rating-setting`): Define rating thresholds. Installers falling below 3.5 stars are automatically restricted from receiving Commercial projects.
4. **Installer Agency Plans** (`/admin/settings/installer/agency-plans`): Paid subscription tiers for third-party installation companies.

3.9.3 C. Backend API Endpoints & Mongoose Models

Models: `Installer.js`, `ToolRequirement.js`, `RatingRule.js` (solar-server/models/partner/ & settings/)

- **PUT** `/api/installers/:id/rating` — Updates cumulative average star rating after customer inspection sign-off.

3.10 Module 10: Product Configuration Module

3.10.1 A. Business Purpose

Establishes the core catalogue taxonomy for SOLARKITS. Defines project types (Residential / Commercial / Agricultural), unit measurements (kW, Watt, Metres), and master price catalogues.

3.10.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Product Configuration (Expandable menu)

1. **Add Project Type** (/admin/settings/product/add-project-type): Define master project classifications (Residential Rooftop, Ground Mount, Solar Pump).
2. **Add Project Category** (/admin/settings/product/add-project-category): Sub-categorise by capacity slabs (e.g., 1 kW–5 kW, 5 kW–10 kW).
3. **Add Product** (/admin/settings/product/add-product): Master inventory hardware item setup.
4. **SKU** (/admin/settings/product/sku): Automated SKU code generator. Select Category, Capacity, and Brand to generate standardised codes (e.g., INV-5KW-GRO-01).
5. **Price Master** (/admin/settings/product/price-master): Set wholesale dealer base prices and retail customer MSRP per Watt/kW.
6. **Add Unit Management** (/admin/settings/product/add-unit-management): Define metric units (Pieces, Rolls, Sets, kW).

3.10.3 C. Backend API Endpoints & Mongoose Models

Models: ProductCategory.js, ProductItem.js, PriceMaster.js (solar-server/models/settings/)

- **POST** /api/products/sku-generate — Executes deterministic SKU generation algorithm, checks for collisions, commits product master record.

3.11 Module 11: ComboKit & AMC Module

ComboKit Architecture Matrix

Master Component	Included Hardware Bundle
5 kW Residential Kit (Standard) SKU:CK-5KW-RES-STD	• 10× 500 W Bifacial Solar Panels • 1× 5 kW String Inverter (Wi-Fi) • 50 m DC Wire + AC/DC Combiner Box • GI Mounting Structure (Standard Roof)

3.11.1 A. Business Purpose

Combines standalone products into ready-to-install pre-packaged Solar Kits (e.g., “5 kW Premium Kit: 10× 500 W Panels + 5 kW Hybrid Inverter + Mounting Structure”). Also manages Annual Maintenance Contract (AMC) plans.

3.11.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → ComboKit (Expandable menu)

1. **Create Solarkit** (/admin/settings/combokit/create-solarkit): Kit builder UI. Select capacity (10 kW) and assign precise panel, inverter, and balance-of-system

(BOS) quantities.

2. **Create AMC Plans** (/admin/settings/combokit/create-amc): Define 1-year, 3-year, and 5-year maintenance contract pricing.
3. **AMC Services** (/admin/settings/combokit/amc-services): List of covered maintenance activities (panel cleaning, inverter firmware updates).
4. **Solarkit Bundle Plans** (/admin/settings/combokit/bundle-plans): Multi-kit promotional bundle configurations.
5. **Add ComboKit** (/admin/settings/combokit/add-combokit): Standard single kit creation modal.
6. **Customize Combokit** (/admin/settings/combokit/customize): Configure swap rules. Allows dealers to upgrade inverters during project sign-up subject to real-time price differentials.
7. **Combokit Overview** (/admin/settings/combokit-overview): Master inventory summary of all active pre-packaged kits.

3.11.3 C. Backend API Endpoints & Mongoose Models

Models: ComboKit.js, AmcPlan.js (solar-server/models/settings/)

- **GET** /api/combokits/calculate-price — Sums individual component master prices, subtracts bundled discount percentage, returns final net pricing.

3.12 Module 12: Partner Settings Module

3.12.1 A. Business Purpose

Manages external sales networks: Franchisees, Dealers, and Lead Generation Partners. Configures onboarding targets, subscription plans, and commission/reward point structures.

3.12.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Partner Settings (Expandable menu)

1. **Partner Plans** (/admin/settings/partner/plans): Configure partnership subscription fee tiers (e.g., “Premium Dealer Plan — Rs.25,000/yr”).
2. **Partner Points & Reward Setting** (/admin/settings/partner/points-rewards): Setup cashback rules: *1 kW Installed = 100 Reward Points (Value Rs. 1,000)*.
3. **Partner Onboarding Goals** (/admin/settings/partner/onboarding-goals): Assign monthly lead generation quotas to regional Franchisee Managers.
4. **Partner Profession Type** (/admin/settings/partner/profession-type): Classify partners (Architect, Electrician, Real Estate Broker).
5. **Add Partner** (/admin/settings/partner/add-partner): Direct partner onboarding KYC form.

6. **Partner Manager Setting** (/admin/settings/partner-manager): Configure hierarchy rules for internal managers overseeing partner networks.
7. **Partner Buy Lead Setting** (/admin/settings/partner-buy-lead): Set pricing for unassigned company leads purchased by franchisees.

3.12.3 C. Backend API Endpoints & Mongoose Models

Models: PartnerPlan.js, PartnerPoints.js, OnboardingGoal.js (solar-server/models/partner/)

- **POST** /api/partners/credit-points — Triggered automatically when Project Stage 8 (Completion) is signed off; adds reward points to dealer wallet.

3.13 Module 13: Quotes & BOM (Bill of Materials) Module

3.13.1 A. Business Purpose

Automates customised commercial and residential solar quotation generation. Calculates shadow-free roof terrace area, structural wind load requirements, and local Discom net-metering charges.

3.13.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Quote (Expandable menu)

1. **Quote Setting** (/admin/settings/quote/quote-setting): Define quotation PDF terms and conditions, validity periods (30 days), and tax headings.
2. **Survey BOM Setting** (/admin/settings/quote/survey-bom-setting): Configure rules linking survey measurements to component quantities.
3. **Terrace Setting** (/admin/settings/quote/terrace-setting): Set shadow clearance coefficients (e.g., 100 sq ft shadow-free area required per 1 kW).
4. **Structure Setting** (/admin/settings/quote/structure-setting): Configure mounting structure parameters (Standard RCC roof vs Tin shed vs High-rise wind loads).
5. **Building Setting** (/admin/settings/quote/building-setting): Floor height wiring length estimators.
6. **Discom Master** (/admin/settings/quote/discom-master): Table of state electricity board net-metering application fees (e.g., DGVCL Gujarat = Rs. 3,500).

3.13.3 C. Backend API Endpoints & Mongoose Models

Models: Quotation.js, DiscomMaster.js, StructureConfig.js (solar-server/models/finance/ & settings/)

- **POST** /api/quotations/generate — Receives survey dimensions, evaluates structural load, calculates Discom fee, returns formatted PDF data stream.

3.14 Module 14: Delivery & Logistics Module

3.14.1 A. Business Purpose

Manages the transportation of heavy solar equipment from warehouses to customer installation sites. Configures delivery vehicle capacities, transport routes, and freight charges.

3.14.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Settings Operations → Delivery Settings
(Expandable menu)

1. **Delivery Type** (/admin/settings/delivery/delivery-type): Define transport modes (Standard Trucking, Express Courier, Self-Pickup).
2. **Vehicle Selection** (/admin/settings/delivery/vehicle-selection): Table of transport trucks (Tata Ace, Eicher 14ft). Form: Weight limit (Tons) and volumetric space. The system prevents dispatch if the project ComboKit weight exceeds truck limits.
3. **Vendor Delivery Plan** (/admin/settings/delivery/vendor-delivery-plan): Configure distance-based freight contracts (e.g., Rs. 50/km for distances > 50 km).

3.14.3 C. Backend API Endpoints & Mongoose Models

Models: DeliveryVehicle.js, DeliveryPlan.js (solar-server/models/vendors/ & settings/)

- **POST** /api/delivery/calculate-freight — Evaluates delivery GPS distance and total kit weight, matches against DeliveryPlan slabs, calculates transport cost.

3.15 Module 15: Loan & Finance Configuration Module

3.15.1 A. Business Purpose

Integrates solar financing rules. Connects customer loan approvals with project construction milestones (e.g., 20% disbursement on Stage 3, 60% on Stage 5, 20% on Stage 8).

3.15.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Loan Setting (/admin/settings/loan)

1. **Loan Rule Builder:** Multi-select form to select eligible **Product Categories** (ComboKits > 5 kW) and **Partner Types** (Franchisees).
2. **Financial Parameters:** Set minimum down-payment percentage (15%) and minimum CIBIL score (650).
3. **Disbursement Schedule:** Define milestone disbursement percentages.

Related Dashboard: Sidebar → Dashboard → Orders by Loan (/admin/dashboard/orders-by-loan) — Tracks live loan applications across partner banks.

3.15.3 C. Backend API Endpoints & Mongoose Models

Model: LoanRule.js (solar-server/models/finance/LoanRule.js)

- **PUT** /api/loans/update-disbursement — Listens for project stage progression webhooks, updates bank disbursement status in MongoDB.

3.16 Module 16: Sales & Marketing Module

3.16.1 A. Business Purpose

Manages pricing markups, promotional discount coupons, AMC contract renewals, and lead generation ad campaigns across Google/Meta ad networks.

3.16.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Sales Settings / Marketing Settings

1. **Set Price** (/admin/settings/sales/set-price): Define state-wise pricing markups. Escalated special price requests are flagged with an active red notification badge.
2. **Set Price For AMC** (/admin/settings/sales/set-price-amc): AMC contract pricing slabs.
3. **Offers** (/admin/settings/sales/offers): Promotional discount coupon codes (e.g., SUMMER15 for Rs. 15,000 off). Form: Discount value, expiry date, and minimum kW eligibility.
4. **Solar Panel Bundle Setting** (/admin/settings/sales/solar-panel-bundle-setting): Multi-panel wholesale bundle markups.
5. **Campaign Management** (/admin/settings/marketing/campaign-management): Track Google/Meta advertising budgets, UTM parameters, and generated lead conversion counts.

3.16.3 C. Backend API Endpoints & Mongoose Models

Models: Campaign.js, Offer.js, PriceMarkup.js (solar-server/models/marketing/ & settings/)

- **POST** /api/marketing/leads/webhook — Captures incoming lead from Facebook/Google ads, matches UTM code to Campaign, assigns lead to regional Franchisee.

3.17 Module 17: Checklist & SOP (Standard Operating Procedures) Module

3.17.1 A. Business Purpose

Enforces strict quality control by establishing mandatory verification checklists for every project stage. Prevents safety violations or substandard work by field installers.

3.17.2 B. UI Navigation & Sub-sections

Path: Sidebar → Settings → Checklist Setting (/admin/settings/checklist)

1. **SOP Template Builder:** Select Project Stage (e.g., Stage 6 — Installation). Add verification items (e.g., “Check earthing pit resistance < 5 Ohms”). Check the “**Photo Mandatory**” box for critical items such as Inverter Wiring.
2. **Field Execution:** When installers update progress from their mobile app, they cannot advance the project stage until all checklist items are ticked and required photos are uploaded.

3.17.3 C. Backend API Endpoints & Mongoose Models

Models: ChecklistTemplate.js, ProjectChecklist.js (solar-server/models/settings/ & projects/)

- **GET** /api/checklists/stage/:stageKey — Returns required checklist items for mobile UI rendering.

3.18 Module 18: Analytics & Reports Module

Financial Analytics Dashboard

Report Category	Key Performance Indicators (KPIs)
Financial & P&L	Gross Revenue, EBITDA, Net Margin %
Cashflow Statement	Inflow (Collections), Outflow (Vendor)
Inventory Valuation	Total Stock Value (FIFO/LIFO basis)
Regional Revenue Breakdown	Revenue by State, District, & Cluster

3.18.1 A. Business Purpose

Provides real-time business intelligence and financial reporting. Generates board-ready financial statements, inventory valuations, captable summaries, and regional revenue maps.

3.18.2 B. UI Navigation & Sub-sections

Path: Sidebar → Report (Expandable menu)

1. **Financial & P&L** (/admin/reports/financial-pl): Gross revenue, direct component costs, dealer commissions, and net profit margins. Filters by Date range and Region.
2. **Cashflow** (/admin/reports/cashflow): Inflows (Customer milestones) vs Outflows (Vendor payouts, logistics freight).
3. **Inventory** (/admin/reports/inventory): Total warehouse stock valuation calculated on FIFO/LIFO basis.
4. **Loans** (/admin/reports/loans-summary): Bank-wise financed order totals and pending disbursement totals.
5. **Captable** (/admin/reports/captable): Shareholding and equity distribution records.
6. **Revenue By CP Types** (/admin/reports/revenue-by-cp-types): Channel partner sales contribution comparison (Franchisees vs Dealers vs Direct Sales).
7. **Cluster, District, & City Reports** (.../cluster, .../district, .../city): Granular geographical revenue maps. All report views include an **Export to Excel / PDF** button for external accounting audits.

3.18.3 C. Backend API Endpoints & Mongoose Models

Models: Aggregates data dynamically across Project, Quotation, InventoryItem, and Vendor collections.

- **GET** /api/reports/financial-summary — Executes complex Mongoose aggregation pipelines (\$group, \$match, \$lookup) across multiple financial tables; returns formatted analytical datasets.

4. Enterprise Security, Authentication & Access Control

SOLARKITS ERP implements strict security protocols to safeguard sensitive customer KYC, financial transactions, and operational data.

Role-Based Access Delegation

User Role	System Permission Scope
Super Admin	Full read/write access across all 18 Master Modules.
HR Manager	Access limited to Departments, HRMS, & User Roles.
Warehouse Incharge	Access limited to Operations, Inventory, & Dispatch.
Dealer / Franchisee	Access limited to assigned regional project leads.

4.1 Technical Implementation

- **JWT Authentication:** Handled via custom Express middleware (`protect`). Validates Bearer token headers against active user sessions.
- **Role Authorization:** Handled via `authorize('admin', 'dealer')`. Restricts API execution based on user role designations.
- **Department Delegation:** In `AdminSidebar.jsx`, the UI dynamically evaluates `user.department.assignedModules` and `user.delegatedDepartments` to hide or show specific sidebar menu links.

5. Production Readiness & Client Verification Summary

The SOLARKITS ERP Admin Panel is fully built, professionally styled, and successfully deployed in the staging environment.

5.1 Current System Status

- **Frontend UI/UX:** All 18 modules and their nested sub-sections are fully wired with high-fidelity React components, responsive tables, modal forms, and clean vector iconography.
- **Core Data Engine:** Standard CRUD controllers, geographical models, and hierarchical queries are fully operational.
- **Final Audit Checklist Wiring (In Progress):** To complete the final client checklist and achieve 100% green-tick compliance prior to the production GitHub push, the backend database audit logging layer (`ActivityLog` and `DeleteLog` Mongoose middleware) is now being attached to the controllers of all 18 modules.